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Editorial Office
Computer Standards & Interfaces
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Dear Editors,

Re: Submission to *Computer Standards & Interfaces*
"A Machine-Readable Standards-Traceability Framework for Cybersecurity Risk Evidence in IoT Environments"

Please consider the following manuscript for publication as an original research article in *Computer Standards & Interfaces*:

> **"A Machine-Readable Standards-Traceability Framework for Cybersecurity Risk Evidence in IoT Environments"**

Scope fit. The manuscript presents a machine-readable standards-traceability artifact for cybersecurity risk evidence that combines a UML/XMI representation, a Goal–Question–Metric validation layer, and standards-linked downstream decision support. Its primary emphasis is on the specification, validation, and application of standards-aligned representations and interoperable evidence mappings for information-security workflows, which fits the journal's focus on standards, interfaces, information management, and applied methods.

Main contributions.

- A machine-readable UML/XMI and GQM artifact covering **93/93** ISO/IEC 27001:2022 Annex A controls and **106/106** NIST CSF 2.0 subcategories, together with validator scripts and logs.
- A standards-traceable quantitative risk formulation based on AV×EP×IL with explicit provenance for CVSS, asset-value, and optional human-factor inputs.
- A telemetry-only IDS evidence-generation path that excludes downstream enrichment fields from the classifier feature set and uses boundary-safe train/validation/test partitioning on the public CIC IoT 2023 profile.
- A Gordon–Loeb-bounded portfolio comparison layer that exposes explicit cost-risk trade-offs while preserving links back to controls, metrics, and represented evidence objects.

Reported empirical results. On the strict public CIC IoT 2023 profile reported in the manuscript, the artifact-default scorer achieves **F1 = 0.995**, **AUROC = 0.997**, **PR-AUC = 1.000**, and **Brier = 0.006**. At the operating point selected on the pre-test window for a nominal 1% FPR target, the hold-out test window reports approximately **98.2%** detection rate at **0.44%** realized FPR. On the optimization side, NSGA-II returns a **61**-point frontier (HV=**31,436.9**), Greedy returns a **14**-point prefix frontier (HV=**30,810.2**), and Exact-ILP returns a **16**-point frontier (HV=**28,970.3**). The artifact-default NSGA-II selection yields **4.5%** lower residual risk at **5.0%** higher three-year control-side TCO than the matched-budget Greedy comparator, making the trade-off explicit.

Reproducibility and limitations. The submission includes code, schema artifacts, mappings, and validator outputs. Public source and manuscript materials are available through the project repository at GitHub. The manuscript is explicit about the main public-data limitations: victim/device IDs are not exposed in the public CIC IoT 2023 CSV rows, asset values are telemetry proxies rather than enterprise register values, and the ordered hold-out in the merged CSV release remains a surrogate for true wall-clock chronology because row-level timestamps are absent.

Submission package. The manuscript is accompanied by a separate highlights file and a separate graphical abstract file, aligned with the current *Computer Standards & Interfaces* Guide for Authors.

Declarations.

- The manuscript is original, unpublished, and not under consideration elsewhere.
- The study uses public or non-identifiable security telemetry and vulnerability data; no human-subjects experiment was conducted.
- There is no external funding and no conflict of interest to declare.
- The accompanying artifact is publicly available through the GitHub repository cited above.

Thank you for your consideration.

Sincerely,

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encl: Manuscript; Figures and tables; Supplementary materials